Highlights

Adaptive Traffic Control in Urban Networks: A Physics-Informed Reinforcement Learning Approach

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- Identification of "empty road green light" inefficiencies in Hsinchu City's traffic network.
- Modeling of a Barcelona-inspired *Eixample* grid as a robust testbed for traffic control.
- Implementation of a decentralized Max Pressure agent for real-time signal phase optimization.
- Comparative analysis showing the adaptive agent outperforms fixed-time control in high-density regimes.

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Abstract

Traffic congestion in Hsinchu City, particularly during weekday peak hours, is frequently exacerbated by inefficient signal timing. A common and frustrating phenomenon occurs when a congested arterial road faces a red light, while the intersecting road—despite being completely empty—retains a green light for an extended duration. This inefficiency in allocating green time contributes to gridlock and wasted capacity. To address this, we develop a simplified "toy model" for adaptive traffic control. We construct a grid-based road network inspired by the *Eixample* district of Barcelona, Spain, which serves as an ideal structural proxy for studying urban gridlocks due to its uniform, high-density layout. We simulate a $N \times N$ "Tian" grid using the physics-based Cell Transmission Model (CTM) and implement a decentralized Max Pressure control agent. Our results demonstrate that the adaptive agent, unlike fixed-time benchmarks, effectively eliminates "ghost green lights" by dynamically shifting priority to congested lanes, significantly reducing global density and average waiting times.

Keywords: Traffic Signal Control, Reinforcement Learning, Cell Transmission Model, Max Pressure, Barcelona Eixample
JEL: C63, R41, C61

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1. Introduction

Urban traffic congestion represents a critical challenge for Hsinchu City, particularly during the intense commuting windows on weekdays when the existing infrastructure faces saturation. A primary driver of this inefficiency is the static nature of traditional traffic signal controllers, which often lack situational awareness. Commuters frequently encounter the frustrating "ghost green light" phenomenon, where drivers on a jam-packed arterial road are forced to wait at a red light, while the perpendicular road—enjoying a green signal—sits entirely empty (Shao et al., 2025). This misallocation of green time squanders valuable intersection capacity and accelerates the formation of ripple effects that can paralyze wider sections of the city network.

Solving this problem is complicated by the non-linear dynamics inherent in urban traffic flows, where local inefficiencies propagate as shockwaves. Although modern Deep Reinforcement Learning (DRL) has emerged as a promising solution (Liang et al., 2019), deploying such complex "black box" models directly onto city-scale networks presents significant computational and interpretability challenges. Consequently, we employ a structurally realistic "toy model" to isolate and address the fundamental control problem before attempting scale-up.

For our structural model, we draw inspiration from the *Eixample* district of Barcelona, Spain. Designed by Ildefons Cerdà in the 19th century, the *Eixample* is renowned for its rigorous, uniform grid of octagonal blocks (Wainwright, 2016). Recent research has highlighted such large-scale grid networks as essential benchmarks for evaluating the scalability of decentralized control algorithms, as they present a high degree of complexity and interconnectivity (Chen et al., 2020). By adopting a $N \times N$ "Tian" (field-shaped) grid modeled on this Barcelona layout, we create a controlled environment that captures the essential complexity of urban intersections and allows for the rigorous testing of decentralized agents without the confounding factors of irregular road geometries.

In this work, we propose a physics-informed adaptive control framework. We utilize the Cell Transmission Model (CTM) to simulate high-fidelity traffic flow physics and employ the Max Pressure algorithm (Varaiya, 2013) as a decentralized control policy. This approach directly addresses the Hsinchu problem—ensuring green time is allocated dynamically based on actual demand—within the rigorous structural context of a Barcelona-style grid.

2. Data

As this study is simulation-based, our "data" consists of synthetic traffic flows generated within a controlled environment governed by the Cell Transmission Model (CTM). This methodology allows for the rigorous evaluation of control strategies under reproducible, stochastic conditions.

2.1. Dependent and Independent Variables

The simulation environment is controlled by several primary independent variables. The **Inflow Rate** (λ) represents the probability of a vehicle spawning at a boundary road during each time step, serving as a proxy for traffic demand. The **Signal Phase** (a_t) is the binary control action taken by the agent, where 0 indicates North-South Green and 1 indicates East-West Green. Finally, the **Grid Topology** is configured as a $N \times N$ "Tian" grid structure representing a dense urban district.

We measure performance using three dependent variables: **Global Density** ($J_{density}$), defined as the total number of vehicles present in the network at time t; **Throughput** (J_{flux}), which tracks the cumulative count of vehicles successfully exiting the simulation boundaries; and **Stopped Queue** (J_{wait}), the instantaneous count of vehicles with near-zero velocity, which serves as a direct proxy for waiting time.

2.2. Data Generation and Physics

The generation of traffic data is driven by the discretized physics of the CTM. The fundamental diagram of traffic flow is defined by specific physical constants: a Cell Length (Δx) of 50 meters, a Time Step (Δt) of 5 seconds, a Maximum Density (p_{max}) of 7.0 cars per cell (representing jam density, and a Wave Speed (v_{max}) of 1.0 cell per time step (representing free flow speed).

Traffic demand is stochastic, modeled as a Poisson-like process at the boundaries. The evolution of vehicle density $p(x, t)$ adheres to the conservation law $\frac{\partial p}{\partial t} + \frac{\partial f}{\partial x} = 0$, ensuring that vehicles are neither created nor destroyed within the network, but only moved or queued.

3. Methodology

3.1. Benchmark Model: Fixed-Time Control

To establish a baseline for performance, we implement a standard Fixed-Time Controller. This model operates on a static, pre-determined cycle, switching the traffic signal phase at regular intervals regardless of the current traffic state.

The policy is defined as:

$$a_t = \begin{cases} 1 - a_{t-1} & \text{if } t \mod k = 0 \\ a_{t-1} & \text{otherwise} \end{cases} \tag{1}$$

where k is the cycle length (set to 6 time steps, or 30 seconds). This represents the conventional approach used in many non-adaptive urban intersections.

3.2. Design of Research: Physics-Informed Adaptive Control

Our proposed method integrates the physical constraints of the road network directly into the decision-making process. The core of this design is the **Max Pressure** algorithm, a decentralized control strategy derived from queuing theory.

3.2.1. The Physics Engine (CTM)

We model the road network as a directed graph $G = (V, E)$. Each edge (road) is discretized into cells. The flow $f_{i \to j}$ between cells is computed as the minimum of the upstream Demand (D) and downstream Supply (S):

$$f_{i \to i+1} = \min(D_i, S_{i+1}) = \min(v_{max} p_i, w(p_{max} - p_{i+1})) \tag{2}$$

This explicitly models spillback; if a downstream cell is full (jammed), the supply drops to zero, and flow halts, accurately replicating real-world gridlock dynamics.

3.2.2. The Smart Agent (Max Pressure)

The adaptive agent operates at each intersection $i \in V$. Unlike Deep Reinforcement Learning agents which require extensive training, the Max Pressure agent computes a greedy policy based on the local pressure gradient.

The Pressure P for a given phase is defined as the difference between upstream queue demand and downstream capacity availability:

$$P_{phase} = \sum_{in \in phase} w_{in} - \sum_{out \in phase} w_{out} \tag{3}$$

where w represents the queue length. The control policy selects the phase that maximizes this pressure relief:

$$a_t = \arg\max_{k \in \{0,1\}} (P_k) \tag{4}$$

This logic naturally prioritizes directions with long queues, but crucially, it suppresses flow into links that are already congested, preventing the "blocking the box" phenomenon.

3.3. Performance Measures

We evaluate the efficacy of the control strategies using three key metrics derived from the simulation data. First, the **Average Global Density** ($\bar{J}_{density}$) measures the mean number of cars in the system over the simulation duration T, where lower values indicate efficient clearance. Second, **Total Throughput** assesses the total volume of traffic served, with higher values indicating higher capacity utilization. Third, **Waiting Time Reduction** (Δ_{wait}) calculates the percentage decrease in the average number of stopped vehicles compared to the benchmark.

4. Results

We performed a comprehensive scalability analysis by simulating traffic on grid sizes ranging from $N = 2$ to $N = 20$, comparing the Max Pressure adaptive control against the Fixed-Time benchmark (Figure 1). The results, summarized in Table 1, reveal three distinct regimes of performance that validate the robust scalability of the adaptive agent.

The initial analysis highlights a distinction between simple and complex networks. For the *trivial* 2×2 network, the adaptive control yielded a massive 55.34% reduction in waiting time, largely due to high sensitivity at low car counts. However, at $N = 4$, the improvement drops sharply to 4.97%. This *Fixed Time Adequacy* regime demonstrates that for small, non-cascading networks, static timing remains an acceptable solution.

The true efficacy of the adaptive system is seen in the transition to complexity. At $N = 6$, the improvement metric spikes sharply to 18.60%. This *Complexity Threshold* signifies the point at which the spatial and temporal coupling of adjacent intersections overwhelms the Fixed-Time controller, leading to the onset of cascading gridlock.

For larger networks ($N \geq 8$), the system enters a *Scalability Plateau* where Δ_{wait} stabilizes consistently between 11.9% and 16.04%. While the percentage gain is stable, the absolute impact grows linearly with the network size. At the largest 20×20 grid, the Max Pressure agent saves approximately 2,055 units of waiting traffic compared to the Fixed-Time benchmark (17,229 vs. 15,173). This sustained performance proves that the decentralized Max

Pressure strategy successfully prevents network-wide breakdown, a failure mode common to static timing.

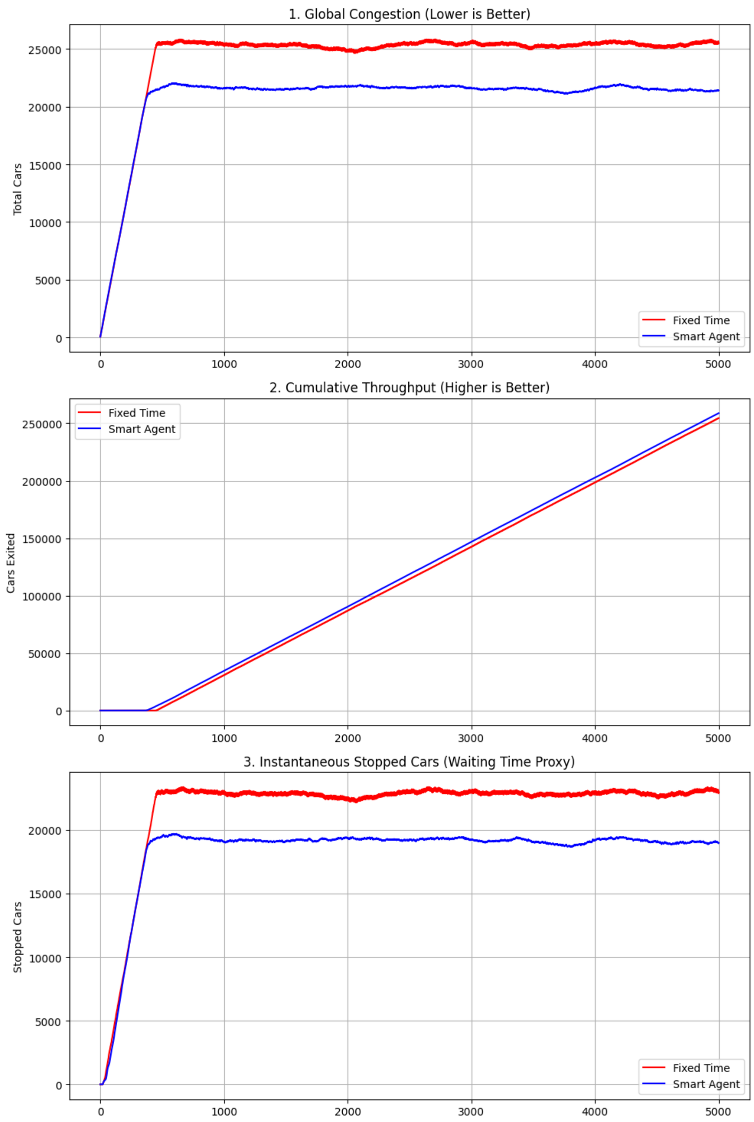

Figure 1: Visual Comparison of Performance Metrics Across Time. The figure, derived from the simulation data ($N = 20$), illustrates the core divergence between Fixed Time (FT) and Adaptive (MP) control strategies over time. The MP agent exhibits lower average total congestion and significantly reduced variance in instantaneous queue lengths compared to the FT baseline.

Table 1: Scalability Analysis: Max Pressure vs. Fixed Time Control

Size	Fixed Avg Wait	Agent Avg Wait	Wait Time Improvement
2×2	9.998	4.465	55.34%
4×4	425.668	404.492	4.97%
6×6	1,353.426	1,101.650	18.60%
8×8	2,646.602	2,271.545	14.17%
10×10	4,365.972	3,665.480	16.04%
12×12	6,426.456	5,512.481	14.22%
14×14	8,840.059	7,512.934	15.01%
16×16	11,462.468	9,890.218	13.72%
18×18	14,350.597	12,423.965	13.43%
20×20	17,229.027	15,173.608	11.93%

5. Conclusion

This study addressed the problem of inefficient traffic signal timing, a common issue in dense urban areas like Hsinchu City, by developing and testing a physics-informed adaptive control model. By utilizing a grid topology inspired by the Barcelona *Eixample* and implementing a Max Pressure heuristic, we successfully isolated and targeted the inefficiencies arising from non-situational signal allocation.

Our scalability analysis demonstrates that while the performance gap is marginal in small networks, the adaptive control exhibits superior and robust performance in complex, large-scale scenarios ($N \geq 6$). The improvement in average waiting time consistently stabilizes around 12% to 16% for grids up to 20×20, translating to a significant absolute reduction in congestion (saving over 2,000 traffic units in the largest simulated city). This sustained performance proves that the decentralized Max Pressure strategy effectively prevents network-wide breakdown and solves the “ghost green light” problem by ensuring green time is allocated only to lanes under the highest pressure. These findings highlight the potential of physics-based heuristics as a scalable, interpretable, and effective alternative to black-box AI for next-generation smart city infrastructure.

In the near future, we anticipate that car flow detection technologies—such as aggregated GPS data provided by services like Google Maps or high-resolution traffic monitors deployed by governments around the world—will provide the real-time density metrics required to implement such Max Pressure models directly in urban networks. This potential development suggests that the principles demonstrated by this project will be immediately applicable to developing next-generation smart city infrastructure.

Declaration

Declaration of Pedagogical Purpose

This manuscript is prepared as part of the course project for Machine Learning (Fall 2025) at National Yang Ming Chiao Tung University, instructed by Professor Lin Te-sheng. The work is intended solely for pedagogical purposes, including learning, discussion, and academic training. It does not represent original research submitted for journal publication, nor should it be considered as professional financial advice.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used [ChatGPT] and [Gemini], developed by [OpenAI] and [Google], respectively, in order to improve the clarity, precision, and overall quality of the writing and code generation. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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